

MATT JAEHN - Full Stack Engineer

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PROFESSIONAL SUMMARY

Results-driven Full Stack Engineer with over a decade of experience designing and implementing enterprise-scale solutions. Proven track record of leading cross-functional teams, modernizing legacy systems, driving greenfield projects, and delivering performance-critical applications. Expertise spans cloud architecture, data systems, and full-stack development with a strong foundation in computing theory and mathematics.

CORE COMPETENCIES

Languages & Frameworks

- JavaScript/TypeScript, Node.js, React (Next.js, Redux, RTK)
- Java, Java Bootstrap, Spark
- Golang, Python, Django, C#/.NET (F#, C++), Java/Groovy, C/C++
- Elixir, PL/SQL, T-SQL, Haskell, Clojure, Scala, Rust

Cloud & Infrastructure

- AWS Data Engineering Associate
- AWS (CDK, CloudFormation, ECS, Lambda, S3, DynamoDB, Kinesis)
- Google Cloud Platform, Microsoft Azure
- Infrastructure as Code, DevOps, CI/CD

Databases & Data Systems

- Oracle RDBMS, SQL Server, PostgreSQL, MySQL/MariaDB
 - Extensive experience in optimizing queries and stored procedures
 - Extensive experience writing stored procedures in PL/SQL and T-SQL
 - MongoDB, DynamoDB, DocumentDB, Neo4j
 - Writing extension modules for Neo4j (Java)
 - Querying Neo4j using Cypher, Gremlin, as well as SDKs
 - Optimization of DynamoDB indexing schemes
 - leveraging atomic guarantees in Mongo/DocumentDB and DynamoDB to implement consistency invariants in applications
 - ETL Pipelines, Data Architecture, GraphQL
 - Lucene/ElasticSearch indexes
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PROFESSIONAL EXPERIENCE

Data Architect | *Orion Federal Credit Union* | Memphis, TN

September 2023 - May 2025

- Architected unified data API serving as single interface to core banking systems, improving system integration and reducing development complexity
- Researched and Prototyped OCR/Computer Vision functionality for check and customer identity documents capture, evaluating various neural nets and image processing techniques
- Led CDK-first infrastructure design implementing GraphQL backend with comprehensive AWS services including ECS clusters, dockerized Lambda functions, RDS, DynamoDB, and Step Functions orchestration.
Over \$3,000,000 in loans have been processed through this API.

- Developed webhook integration system using CDK-managed infrastructure stack enabling seamless third-party vendor integrations and real-time data synchronization
- Designed modern ETL pipelines leveraging AWS DMS for data migration, Step Functions for workflow orchestration, and implemented GraphQL federation architecture for scalable API management

Senior Software Engineer - Manufacturing | *uLab Systems LLC* | Memphis, TN

May 2022 - May 2023

- Bootstrapped manufacturing automation for robotized factory startup, integrating custom machinery with cloud-based web services
- Architected machine-to-cloud integration solutions, designing adapter patterns to bridge proprietary machine interfaces with standardized domain objects
- Delivered 4 production applications handling the assembling of orders and printing of labels compliant with regulations for medical devices
- Built multi-technology stack including Golang/Python cloud services (AWS ECS), React/Redux UIs, and C++ factory floor agents
- Specialized in complex integrations for custom-built machines with limited automation capabilities, creating robust control systems

Lead Developer | *AutoZone* | Memphis, TN

March 2019 - April 2022

- Led search modernization initiative for 6,000+ retail locations as part of company-wide Retail Acceleration program
- Architected distributed replenishment system migrating from per-store local processing to centralized cloud architecture, improving efficiency and reducing maintenance overhead
- Managed development teams of 14-20 engineers across 3-4 sub-teams, overseeing full project lifecycle from analysis through deployment
- Drove performance optimization efforts, implementing tuning strategies that improved system response times by 60% under peak loads
- Delivered enterprise-wide rollout successfully deploying solutions across entire store network without service interruption

Software Developer | *ServiceMaster* | Memphis, TN

December 2014 - February 2019

- Pioneered cloud-first architecture designing and implementing serv-a.io, company's first cloud-deployed RESTful API using AWS, Node.js, and MongoDB
- Built integration infrastructure creating first webhook system for external service integration (Twilio, SendGrid) and internal application connectivity
- Developed file storage API implementing scalable solution using AWS S3, DynamoDB, Lambda, and Kinesis for enterprise document management
- Engineered data migration solution processing 100TB of legacy data through streaming JSON parsing

Software Developer | *Accredo Health, Inc* | Memphis, TN

January 2006 - July 2011

- Automated Oracle Workflow operations developing comprehensive PL/SQL script collection against OWF_MGR schema, allowing for automation of tasks previously requiring manual intervention
- Enhanced operational efficiency providing granular control capabilities beyond Oracle Workflow Manager UI limitations
- Delivered self-directed innovation identifying process improvement opportunities and implementing solutions independently

EDUCATION

Bachelor of Science, Discrete Mathematics
Georgia Institute of Technology | Atlanta, GA

Additional Detail

KEY ACHIEVEMENTS

- Successfully architected and deployed solutions serving 6,000+ retail locations
- Led teams of up to 20 engineers across multiple complex enterprise projects
- Designed first cloud-native solutions for two different Fortune 500 companies

Automated Database Provisioning (CI/CD)

- Using AWS CDK, implemented stacks to provision RDS resources
 - Including both Aurora clusters and bare RDS instances (Postgres)
 - Configuration was handled in a unified way with the overall (version controlled) CDK configuration
 - Database credentials and other secrets implemented as AWS Secrets. Cryptographic keys were handled with AWS KMS.
 - Database initialization (DML, DCL) behavior that could not otherwise be handled through existing CDK/CloudFormation functionality was implemented as custom cloudformation resources, which were then integrated into the overall CDK flow
 - Deployment pipeline was handled through gitlab triggers, fully integrated into the merge request review (when applicable for a given branch). gitlab in turn executed a worker which used the aws cdk CLI to deploy the relevant stacks. These pipeline runs were parameterized by environment with credentials stored in the gitlab secrets functionality.
 - All deployable units were deployed as aws cloudformation stacks, to leverage the monitoring and admin functionality aws provided for stacks.
 - DynamoDB stores were also implemented through CDK, with their schemas integrated into the same CDK configuration. As access to DynamoDB was done entirely through aws IAM, generally no additional secrets were required.
- Using AWS CDK, implemented a database migration solution using Amazon Database Migration Service
 - Again, all components were deployed using CDK (except for the legacy on-prem DB2 database). These components included the compute resources to execute the actual replication, the AWS adapters (called endpoints) for all source and target systems, as well as any required configuration and secrets.
 - Both full load and “real time” on-going CDC replication were implemented.
- Using AWS CDK, implemented and deployed AWS Redshift Clusters
- Using AWS CDK, implemented AWS Athena Glue jobs and related resources as part of ETL pipelines
- Researched a Proof of Concept for the use of Liquibase Database Change Management tool
 - While I’ve never used it in production, I have worked with Liquibase as far as researching its capabilities. It was especially appealing since it could handle Neo4j. However, in the organization I was in at the time, Liquibase ultimately was not ultimately used. This was largely because the main user app at the time used an ORM which implemented DDL version tracking as (ORM-specific) migration files.

Highlighted Proficiencies

- Extensive experience using a variety of Python backend web frameworks
 - Django, including Django’s ORM
 - wsgi server applications
 - directly implemented (without framework)
 - Flask
 - asgi server applications
 - FastAPI
- Extensive experience using Python for:
 - utility CLI programs (“gluing” programs)
 - AWS CDK projects
 - various AWS lambda functions
 - PySpark jobs
 - Computer Vision

- AI Models
 - Image processing (e.g., OpenCV, NumPy)
- Extensive experience using React
 - Redux
 - React Toolkit
 - Chakra UI, Tailwind CSS
 - Nextjs
- CI/CD
 - see [Automated Database Provisioning \(CI/CD\)](#) below